Mahesh Basavaraj

ASST. PROFESSOR · AI RESEARCHER

CONTACT

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GITHUB maheshbasavaraj

YOUTUBE Echoes of AI

SUBSTACK @maheshbasavaraj

AT A GLANCE

16+ years across industry & academia

3 utility patents (2026)

10+ publications & conference papers

Ph.D. in progress · VTU

SKILLS

AI & Machine Learning
Agentic AI · LLMs · Deep Learning · Spatio-Temporal Networks · Reinforcement Learning · NLP · Anomaly Detection · Responsible AI

Software Engineering
Java / J2EE · Python · C++ · Spring Boot · Hibernate · Microservices · SDLC

Data
Python Analytics · MySQL · Oracle · MS SQL Server

Networks & Emerging Tech
IoT QoS Optimisation · Underwater MAC Protocols · Blockchain

DevOps & Tools
Git / GitHub · Jenkins CI/CD · Docker · JIRA

Academic Leadership
Curriculum Design · Outcome-Based Education · FDP Coordination · Industry–Academia Programmes

Bridging production-grade software engineering and frontier AI research — from banking platforms at Infosys and Accenture to Agentic AI in the classroom.

PROFILE

Computer Science educator, AI researcher and former enterprise systems engineer with 8 years building banking and capital-markets software at Infosys and Accenture, and 8 years teaching and researching in engineering academia. Assistant Professor in CSE (Data Science) at DSATM and doctoral researcher under VTU. Holder of 3 utility patents and author of 10+ publications spanning Agentic AI, Large Language Models, spatio-temporal deep learning and cognitive IoT networks. Known for turning research into working systems and for building industry-partnered AI programmes for students and faculty.

ACADEMIC EXPERIENCE

Assistant Professor — CSE (Data Science)
Aug 2024 – Present

Dayananda Sagar Academy of Technology & Management (DSATM)

- Teach and run labs in Data Science, Deep Learning and Agentic AI within an autonomous, VTU-affiliated framework.
- Built and led a flagship 6-month Agentic AI immersion programme for students in partnership with Capabl, alongside institutional Faculty Development Programmes.
- Supervise final-year capstone projects and drive departmental research, including utility patents in machine-assisted automation and IoT systems.

Assistant Professor — CSE
Oct 2023 – Jul 2024

City Engineering College

- Taught core computing courses and supervised applied machine-learning student projects.
- Coordinated departmental technical workshops and student mentoring.

Assistant Professor — CSE
Oct 2021 – Oct 2023

T. John Institute of Technology

- Taught object-oriented programming, data structures and web/cloud architectures.
- Published research evaluating OpenAI Codex on real-world coding and deep-learning credit-card fraud detection.

Lecturer — Computer Science
2018 – 2021

Navya PU College & Janaseva PU College

- Taught pre-university programming, algorithms and computational logic.

INDUSTRY EXPERIENCE

Senior Analyst — Murex Solutions Factory
Feb 2014 – Aug 2017

Accenture

- Developed scalable enterprise modules in Java/J2EE and Spring with relational databases for international capital-markets clients.
- Worked across global agile teams through the full SDLC — JIRA-driven triage, automated builds and release integration.

Senior Systems Engineer · Systems Engineer — Finacle
Jul 2009 – Jan 2014

Infosys

- Built and enhanced Finacle CRM and the Finacle universal banking platform used by banks worldwide.
- Refactored backend services, tuned SQL and transaction-processing paths, and owned system-integration testing and deployments.

EDUCATION

Ph.D., Computer Science & Engineering
DSATM — Visvesvaraya Technological University
Pursuing · enrolled 2023

M.Tech, Computer Science & Engineering
K.S. Institute of Technology — VTU
2021 · CGPA 8.5

B.E., Computer Science & Engineering
UVCE — Bangalore University
2009

CERTIFICATIONS & FDPS

Deep Learning — Elite
NPTEL · 2025

Blockchain Technology
NPTEL · 2025

Data Analytics with Python
NPTEL · 2024

The Science Behind Branding
IISc Bangalore · 2025

FDP: Machine Learning & NLP
Infosys Springboard · 2025

FDP: Cyber Security
AICTE · VTU · 2023

FDP: Amazon Web Services
AICTE · 2022

WRITING & MEDIA

- The Ghost in the Feed: Inside the First AI Civilization
- Thermodynamic Computing: Extropic's Chip and the Path to Efficient AI
- The Patent Paradox: AI and the Future of Intellectual Property

More essays on Substack and AI explainers on the **Echoes of AI** YouTube channel.

PATENTS

01 A Bulk Silkworm Cocoon Inspection and Assisted Sorting Machine Using Shallow-Layer Conveyor Spreading

02 Adaptive Quality of Service Management System for Heterogeneous Internet of Things Networks

03 Ethical and Responsible AI: Bias Mitigation and Fairness in Machine Learning Models

RESEARCH PUBLICATIONS

ICIICS 2026 **Multi-Scale Spatio-Temporal Network with Frequency-Spatial Parallel Attention for Motor Imagery Classification** 3rd Intl. Conf. on Integrated Intelligence and Communication Systems, May 2026

ICIICS 2026 **Predictive Mobility-Aware Cognitive Medium Access Control for Underwater Vehicle Communication** ICIICS 2026

ICBDML 2026 **Harnessing Large Language Models for Next-Generation Artificial Intelligence Applications** 2nd Intl. Conf. on Big Data and Machine Learning, Feb 2026

J. Neurological Sciences **Comment on "Improving care for amyotrophic lateral sclerosis with artificial intelligence and affective computing"** Journal of the Neurological Sciences, Mar 2026

Medical Science Educator **Comment on "Incorporating Artificial Intelligence in Qualitative Research: Exploring the Role of ChatGPT in Thematic Analysis"** Medical Science Educator, 2026

IEEE ICMNWC 2025 **Lightweight Multi-Agent Q-learning Cross-Layer Optimization in Wireless IoT Networks for Balanced Energy Utilization** IEEE ICMNWC, Dec 2025

JETIR / ICEASD 2024 **Using AI Agents Built on LLMs for Building AGI** Intl. Conf. on Engineering Advancements and Sustainable Development, Vol. 11 Issue 5, May 2024

ICAMSnT 2023 **The Alignment Problem in AI** Intl. Conf. on Advances in Modern Science and Technology, Apr 2023

ICAPCTA 2022 **Evaluating OpenAI Codex for Real-World Coding** Intl. Conf. on Advances in Parallel Computing Technologies & Applications, Jan 2022

IRJET 2022 **A Review of Deep Learning Techniques in Detection of Anomaly in Credit Card Transactions** IRJET, Vol. 9 Issue 9

OPEN-SOURCE & TEACHING PROJECTS

github.com/maheshbasavaraj · selected repositories

CPU Rush Node.js · Vanilla JS
Interactive Socratic teaching game for Operating Systems — students predict scheduling, paging, deadlock and file-system behaviour, and a server grades every move.

Data Structure Design Studio Python
Learn data structures by choosing under changing constraints — a progressive, scored decision track from arrays to hash maps.

Game Forge — Educational Game MCP Python · MCP
MCP server that generates single-file browser games, each teaching one concept through play, driven by an editable design doctrine.

ButterCloud Quiz Next.js · Express · Prisma
Scenario-based platform where users pick the best tech stack and architecture for realistic project requirements.

MLOps CI/CD Class Project Python · GitHub Actions
Classroom project: a small production-style ML API with automated tests and model-quality gates on every push.